Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/08/2018
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Genes that Predict the Loss of Bone during Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	11/01/2012	**End Date:**	10/01/2017
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/01/2017 (9/30/17) per NSSC information (Ed., 9/8/16) NOTE: Extended to 10/31/2016 per NSSC information (Ed., 9/10/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae Ph.D. (Jackson Laboratory)		
Grant/Contract No.:	NNX12AL25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The amount of bone loss between individual astronauts is highly variable with some astronauts losing large amounts of bone while others are largely unaffected. This variability is accounted for, at least to a large extent, by genetic variations between individuals. Unfortunately, the specific genes that modulate this genetic sensitivity of bone loss in microgravity are unknown. To this end, we have recently completed a genetic study in which we identified specific regions on chromosomes (QTL or Quantitative Trait Loci) that either add to or reduce the propensity of an individual to lose bone when normal weightbearing is removed from the musculoskeleton. While providing critical information, these QTL still contain hundreds of genes, rather than a single one. Here, we propose first to greatly reduce the number of candidate genes with bioinformatics tools and then experimentally test the shortlist of candidate genes using knockout mice available from the Jackson Laboratory's (JAX) Genetic Resource Repository and the JAX Knock Out Mouse Project (KOMP) production center. The identification of single genes modulating bone loss during weightlessness can be used as a critical diagnostic tool for the identification of astronauts most in need of countermeasures as well as to unravel novel drug targets that preserve bone health in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal directly addresses one of the six highest priority recommendations made in the National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era" (http://www.nap.edu/), namely recommendation AH3: "Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended." Defining some of the specific genes that cause differences in the skeletal sensitivity to weightlessness will not only provide novel diagnostic and therapeutic means of combating bone loss in space but also on Earth in individuals subjected to bedrest or suffering from paralysis.
Task Progress:	In this final no-cost extension period, we attempted to resolve the hurdles that came with breeding tissue-specific LRP4 transgenic mice. While we solved the issues that had us held back, we could not re-start the breeding protocol because funds had been depleted. During the entire funding period, we made significant progress towards the goal of identifying specific genes and pathways that determine the amount of unloading-induced bone loss based on an individual's genetic make-up. We also explored how aging may influence this genetic relationship. Our principal manuscripts resulting from this work can be seen in the Cumulative Bibliography link below.
Bibliography Type:	Description: (Last Updated: 01/16/2018)
Articles in Peer-reviewed Journals	Sankaran JS, Varshney M, Judex S. "Differences in bone structure and unloading-induced bone loss between C57BL/6N and C57BL/6J mice." Mammalian Genome. 2017 Dec;28(11-12):476-86. Epub 2017 Sep 14. https://doi.org/10.1007/s00335-017-9717-4 ; PubMed PMID: 28913652 , Dec-2017
Articles in Peer-reviewed Journals	Jatkar A, Kurland IJ, Judex S. "Diets high in fat or fructose differentially modulate bone health and lipid metabolism." Calcif Tissue Int. 2017 Jan;100(1):20-28. Epub 2016 Nov 10. https://doi.org/10.1007/s00223-016-0205-8 ; PubMed PMID: 27832314; PubMed Central PMCID: PMC5217484 , Jan-2017
Articles in Peer-reviewed Journals	Pongkitwitoon S, Uzer G, Rubin J, Judex S. "Cytoskeletal configuration modulates mechanically induced changes in mesenchymal stem cell osteogenesis, morphology, and stiffness." Sci Rep. 2016 Oct 06;6:34791. https://doi.org/10.1038/srep34791 ; PubMed PMID: 27708389; PubMed Central PMCID: PMC5052530 , Oct-2016
Books/Book Chapters	Wallace IJ, Demes B, Judex S. "Ontogenetic and genetic influences on bone's responsiveness to mechanical signals." in "Building bones: bone formation and development in anthropology." Ed. C.J. Percival, J.T. Richtsmeier. Cambridge: Cambridge University Press, 2017. p. 233-253. (Cambridge Studies in Biological and Evolutionary Anthropology series) https://doi.org/10.1017/9781316388907.011 , Mar-2017